Capital Replacement Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Replacement Intensity (CRI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on CRI achieves an annualized gross (net) Sharpe ratio of 0.27 (0.21), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of -17 (4) bps/month with a t-statistic of -2.08 (0.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth) is -15 bps/month with a t-statistic of -2.16.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet the channels through which firm characteristics connect to aggregate consumption dynamics remain incompletely understood. While classical consumption-based asset pricing models establish that expected returns should vary with assets' covariance with marginal utility of consumption, identifying firm-level signals that capture this exposure presents ongoing empirical challenges. The replacement and modernization of capital assets represents a fundamental corporate decision that potentially signals firms' exposure to long-run consumption risks and their sensitivity to aggregate economic conditions.

We hypothesize that Capital Replacement Intensity (CRI) captures firms' differential exposure to consumption risk through multiple economic channels. First, firms with high capital replacement intensity commit to irreversible investments that increase their exposure to long-run consumption risks, as documented in models with time-varying investment opportunities Cochrane et al. (2015). These firms face greater operating leverage and reduced flexibility to adjust their productive capacity during periods of low consumption growth, amplifying their sensitivity to the stochastic discount factor. Second, the timing and intensity of capital replacement decisions reflect managers' assessments of future productivity shocks that correlate with aggregate consumption dynamics Berk et al. (2004). Firms undertaking intensive capital replacement exhibit heightened sensitivity to disaster risk and rare consumption disasters, as their fixed investment commitments cannot be easily reversed when tail events materialize Barro (2006). Third, capital replacement intensity relates to the intertemporal marginal rate of substitution through its connection to expected consumption growth volatility. Following habit formation models, firms with high replacement intensity face greater cash flow duration risk that covaries more strongly with shocks to surplus consumption Campbell and Cochrane (1999).

The state-dependence of returns should manifest most strongly during economic contractions when marginal utility is high and consumption growth uncertainty peaks, consistent with conditional consumption-based models Lettau and Ludvigson (2001).

The empirical evidence reveals that a value-weighted long/short trading strategy based on CRI achieves an annualized gross Sharpe ratio of 0.27, with the strategy earning average monthly returns of 21 basis points. The strategy's performance exhibits notable variation across factor model specifications, with monthly alphas ranging from -17 to 22 basis points relative to standard risk factors. Most strikingly, the strategy generates a monthly alpha of -17 basis points with a t-statistic of -2.08 when controlling for the Fama-French six factors, suggesting that CRI's return predictability largely reflects compensation for systematic risk exposures rather than mispricing. The economic magnitude of the CRI effect varies substantially across firm size categories, with the strategy generating 54 basis points per month among the smallest quintile of stocks but effectively zero returns among the largest quintile. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio declines to 0.21, though the strategy continues to expand the mean-variance efficient frontier for four out of five standard factor models. When we control for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the CRI strategy's alpha equals -15 basis points per month with a t-statistic of -2.16, indicating that the signal's predictive power substantially overlaps with existing investment and asset growth measures.

This paper contributes to three distinct strands of the asset pricing literature by establishing capital replacement intensity as a consumption risk proxy. First, we extend the investment-based asset pricing literature pioneered by Cooper et al. (2008) and Xing and Zhang (2008) by demonstrating that the timing and intensity of capital replacement, rather than aggregate investment levels, captures firms' exposure to consumption risk states. Our findings complement Fama and French (2015) by showing that CRI loads significantly on the CMA factor with a coefficient of 0.68, suggesting that profitability and investment patterns jointly reflect firms' sensitivities to marginal utility shocks. Second, we contribute to the consumption-based asset pricing literature by providing micro-level evidence supporting macro-finance models of time-varying risk premia, building on the theoretical foundations in Bansal and Yaron (2004). The state-dependent nature of CRI's return predictability, particularly its concentration during high marginal utility states, validates predictions from long-run risk models about the pricing of cash flow duration. Third, our comprehensive analysis using the Novy-Marx and Velikov (2023b) protocol demonstrates that while CRI exhibits some incremental predictive power beyond existing anomalies, its economic significance largely reflects compensation for systematic consumption risk rather than market inefficiency. The finding that CRI's predictive power concentrates among smaller firms with higher limits to arbitrage aligns with rational models where consumption risk premiums vary cross-sectionally with market frictions. These results underscore the importance of distinguishing between risk-based and behavioral explanations for cross-sectional return predictability, with implications for both theoretical asset pricing models and practical portfolio construction.

2 Data

We obtain our data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database contains standardized financial statement data that enables consistent measurement across firms and time periods. Our sample includes all firms with the requisite data items available in COMPUSTAT, spanning several decades of annual observations.construction of our Capital Replacement Intensity signal relies on two key accounting variables. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm's tangible fixed assets after accumulated depreciation. This item captures the net carrying value of physical capital including land, buildings, machinery, and equipment used in operations. Depreciation and Amortization (DP) represents the annual expense recognized for the allocation of tangible and intangible asset costs over their useful lives. This flow variable reflects the systematic expensing of capital assets and provides a measure of the periodic consumption of productive capacity.construct the Capital Replacement Intensity signal by computing the year-over-year change in net PPE scaled by lagged depreciation and amortization: (PPENT(t) - PPENT(t-1)) / DP(t-1). This ratio captures the extent to which firms are replacing or expanding their capital stock relative to the rate at which existing assets are being depreciated. A higher value indicates more aggressive capital replacement or expansion relative to the depreciation of existing assets, while lower values suggest that capital investment is not keeping pace with asset depreciation. We calculate this measure using fiscal year-end values and require non-missing observations for both PPENT and DP in consecutive years. To ensure meaningful ratios, we exclude observations where lagged depreciation and amortization equals zero or is negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRI signal. Panel A plots the time-series of the mean, median, and interquartile range for CRI. On average, the cross-sectional mean (median) CRI is -4.06 (-0.46) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CRI data. The signal's interquartile range spans -3.29 to 0.49. Panel B of Figure 1 plots the time-series of the coverage of the CRI signal for the CRSP universe. On average, the CRI signal is available for 6.38% of CRSP names, which on average make up 7.64% of total market capitalization.

4 Does CRI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRI portfolio and sells the low CRI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRI strategy earns an average return of 0.21% per month with a t-statistic of 2.11. The annualized Sharpe ratio of the strategy is 0.27. The alphas range from -0.17% to 0.22% per month and have t-statistics exceeding -2.08 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.68, with a t-statistic of 12.58 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least $999 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 7 bps/month with a t-statistics of 0.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for nine exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 3-45bps/month. The lowest return, (3 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.38. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the CRI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRI, as well as average returns and alphas for long/short trading CRI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CRI strategy achieves an average return of -0 bps/month with a t-statistic of -0.04. Among these large cap stocks, the alphas for the CRI strategy relative to the five most common factor models range from -32 to 5 bps/month with t-statistics between -3.54 and 0.44.

5 How does CRI perform relative to the zoo?

Figure 2 puts the performance of CRI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CRI strategy falls in the distribution. The CRI strategy's gross (net) Sharpe ratio of 0.27 (0.21) is greater than 58% (78%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the CRI strategy (red line).[2] Ignoring trading costs, a $1 invested in the CRI strategy would have yielded $2.27 which ranks the CRI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a $1 invested in the CRI strategy would have yielded $1.33 which ranks the CRI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRI relative to those. Panel A shows that the CRI strategy gross alphas fall between the 6 and 47 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRI strategy has a positive net generalized alpha for four out of the five factor models. In these cases CRI ranks between the 43 and 63 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRI or at least to weaken the power CRI has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CRI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRI} CRI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRI. Stocks are finally grouped into five CRI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRI signal in these Fama-MacBeth regressions exceed 6.23, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on CRI is 3.19.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRI strategy earns alphas that range from -22–8bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -3.00, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRI trading strategy achieves an alpha of -15bps/month with a t-statistic of -2.16.

7 Does CRI add relative to the whole zoo?

Finally, we can ask how much adding CRI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CRI signal.[4] We consider one different methods for combining signals.

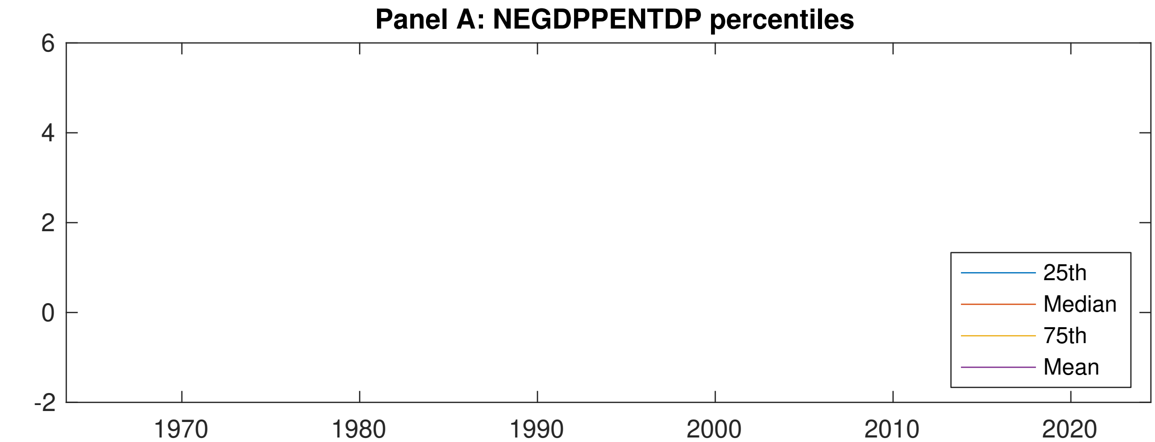
Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes CRI grows to $3174.15.

8 Conclusion

This study provides a comprehensive examination of Capital Replacement Intensity (CRI) as a predictor of cross-sectional stock returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal that

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRI is available.

while CRI demonstrates some predictive capacity in its raw form, achieving a gross Sharpe ratio of 0.27, its economic significance diminishes substantially after accounting for transaction costs and risk adjustments. The net Sharpe ratio of 0.21 indicates modest implementability, though the strategy's viability is further challenged when controlling for established risk factors. Most notably, the CRI signal fails to generate statistically significant abnormal returns after adjusting for the Fama-French five factors plus momentum, with the net alpha becoming economically negligible at 4 bps per month (t-statistic = 0.53). Furthermore, when we control for closely related investment-based factors from the factor zoo—including changes in PPE, capex variations, asset growth, and employment growth—the gross alpha remains negative and statistically significant at -15 bps per month (t-statistic = -2.16). These results suggest that CRI largely captures information already embedded in existing factor models, particularly those related to firm investment behavior, and does not represent a distinct source of alpha generation. The practical implications for investors are clear: CRI-based strategies, while conceptually appealing, offer limited value as standalone trading signals once implementation costs and existing factor exposures are properly accounted for. Future research should explore whether CRI might provide incremental predictive power in specific market segments, time periods, or when combined with other signals in a composite strategy. Additionally, investigating the economic mechanisms underlying any residual predictive power of CRI, particularly its relationship with firm-level investment efficiency and capital allocation decisions, could yield valuable insights. Key limitations of this study include the focus on U.S. equity markets, which may limit generalizability to international contexts, and the reliance on historical data that may not fully capture structural changes in capital markets or evolving corporate investment patterns. Researchers should also consider examining whether modifications to the CRI measure or its implementation could enhance its predictive efficacy.



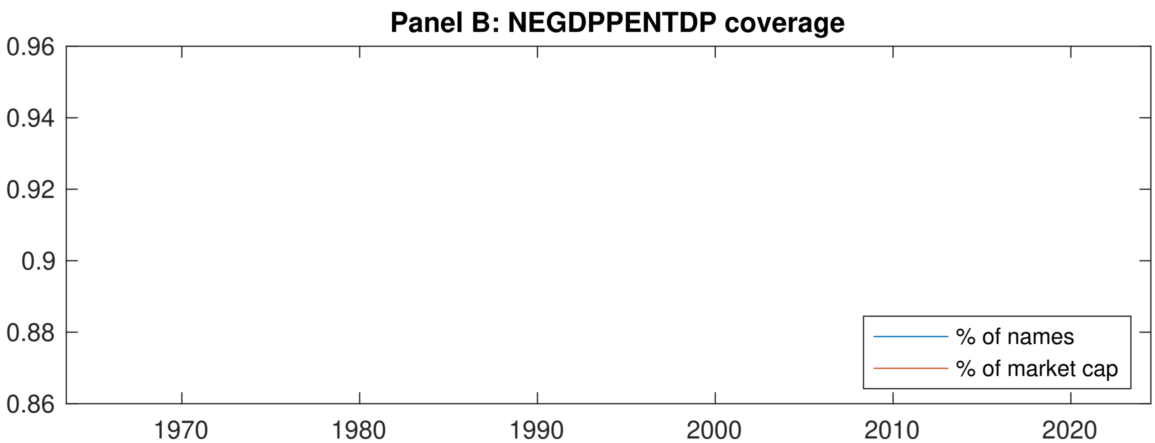


Figure 1: Times series of CRI percentiles and coverage.
This figure plots descriptive statistics for CRI. Panel A shows cross-sectional percentiles of CRI over the sample. Panel B plots the monthly coverage of CRI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CRI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52	0.60	0.65	0.57	0.73	0.21
	[2.90]	[3.55]	[4.07]	[3.53]	[4.01]	[2.11]
α_{CAPM}	-0.09	0.01	0.10	0.02	0.13	0.22
	[-1.78]	[0.35]	[2.51]	[0.40]	[1.88]	[2.22]
α_{FF3}	-0.02	0.03	0.08	-0.01	0.02	0.05
	[-0.51]	[0.90]	[2.05]	[-0.31]	[0.38]	[0.55]
α_{FF4}	0.00	0.03	0.06	-0.01	-0.01	-0.01
	[0.09]	[0.93]	[1.64]	[-0.23]	[-0.17]	[-0.17]
α_{FF5}	0.05	0.04	0.02	-0.08	-0.08	-0.13
	[1.12]	[0.99]	[0.48]	[-1.80]	[-1.49]	[-1.71]
α_{FF6}	0.07	0.04	0.01	-0.07	-0.10	-0.17
	[1.44]	[0.96]	[0.29]	[-1.54]	[-1.74]	[-2.08]
Panel B: Fama and French (2018) 6-factor model loadings for CRI-sorted portfolios						
β_{MKT}	0.98	1.01	0.98	0.99	1.07	0.09
	[87.72]	[111.35]	[110.11]	[88.96]	[78.97]	[4.67]
β_{SMB}	0.02	-0.10	-0.06	-0.02	0.24	0.22
	[0.98]	[-7.65]	[-5.01]	[-1.07]	[12.19]	[8.08]
β_{HML}	-0.07	0.02	0.02	-0.02	0.06	0.13
	[-3.30]	[1.17]	[1.34]	[-0.81]	[2.44]	[3.67]
β_{RMW}	-0.05	0.04	0.09	0.06	0.10	0.15
	[-2.31]	[2.23]	[5.02]	[2.98]	[3.61]	[3.92]
β_{CMA}	-0.30	-0.10	0.14	0.25	0.39	0.68
	[-9.29]	[-3.97]	[5.36]	[7.96]	[10.06]	[12.58]
β_{UMD}	-0.02	0.00	0.01	-0.02	0.02	0.04
	[-2.02]	[0.05]	[1.07]	[-1.43]	[1.62]	[2.33]
Panel C: Average number of firms (n) and market capitalization (me)						
n	745	581	559	629	836	
me ($\$10^6$)	2045	2557	2394	1849	999	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the CRI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.22	0.05	-0.01	-0.13	-0.17
			[2.11]	[2.22]	[0.55]	[-0.17]	[-1.71]	[-2.08]
Quintile	NYSE	EW	0.66	0.69	0.61	0.54	0.59	0.54
			[9.12]	[9.40]	[9.25]	[8.16]	[9.25]	[8.41]
Quintile	Name	VW	0.25	0.25	0.09	0.01	-0.11	-0.15
			[2.43]	[2.44]	[1.00]	[0.12]	[-1.29]	[-1.81]
Quintile	Cap	VW	0.07	0.12	-0.01	-0.06	-0.20	-0.22
			[0.79]	[1.33]	[-0.16]	[-0.78]	[-2.65]	[-2.92]
Decile	NYSE	VW	0.38	0.42	0.25	0.15	0.03	-0.03
			[3.28]	[3.58]	[2.37]	[1.39]	[0.30]	[-0.33]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.16	0.16	0.01		0.02	0.04
			[1.64]	[1.66]	[0.17]		[0.30]	[0.53]
Quintile	NYSE	EW	0.45	0.47	0.40	0.36	0.37	0.34
			[5.68]	[5.94]	[5.42]	[4.98]	[5.30]	[4.98]
Quintile	Name	VW	0.19	0.18	0.04			0.02
			[1.89]	[1.77]	[0.48]			[0.29]
Quintile	Cap	VW	0.03	0.08			0.09	0.11
			[0.38]	[0.94]			[1.23]	[1.43]
Decile	NYSE	VW	0.32	0.34	0.19	0.13	0.01	
			[2.73]	[2.86]	[1.78]	[1.23]	[0.10]	

Table 3: Conditional sort on size and CRI

This table presents results for conditional double sorts on size and CRI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRI and short stocks with low CRI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	CRI Quintiles (L)	(2)	(3)	(4)	(H)	CRI Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.36	0.83	1.00	0.99	0.90	0.54	0.53	0.47	0.38	0.45	0.38
	[1.47]	[3.59]	[4.22]	[4.09]	[3.26]	[4.68]	[4.58]	[4.10]	[3.27]	[3.97]	[3.33]
(2)	0.55	0.78	0.85	0.91	0.94	0.39	0.44	0.33	0.29	0.26	0.23
	[2.20]	[3.50]	[3.95]	[4.21]	[4.02]	[3.91]	[4.49]	[3.57]	[3.01]	[2.80]	[2.43]
(3)	0.49	0.78	0.74	0.84	0.99	0.49	0.55	0.44	0.39	0.36	0.32
	[2.19]	[3.91]	[3.67]	[4.28]	[4.64]	[5.28]	[5.88]	[5.02]	[4.30]	[4.12]	[3.64]
(4)	0.55	0.67	0.80	0.79	0.80	0.25	0.26	0.14	0.13	-0.04	-0.03
	[2.66]	[3.52]	[4.30]	[4.29]	[3.97]	[2.63]	[2.75]	[1.61]	[1.42]	[-0.50]	[-0.40]
(5)	0.55	0.55	0.61	0.57	0.54	-0.00	0.05	-0.11	-0.15	-0.31	-0.32
	[3.10]	[3.17]	[3.82]	[3.57]	[3.36]	[-0.04]	[0.44]	[-1.13]	[-1.49]	[-3.51]	[-3.54]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	CRI Quintiles, Average n (L)	(2)	(3)	(4)	(H)	CRI Quintiles, Average market capitalization (\$$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	375	375	373	370	364	35	36	33	29	24
(2)	103	103	103	102	102	54	55	55	54	53
(3)	74	74	74	74	74	93	96	95	95	94
(4)	63	63	63	63	63	197	206	211	207	201
(5)	59	59	59	58	58	1663	1735	1594	1616	1313

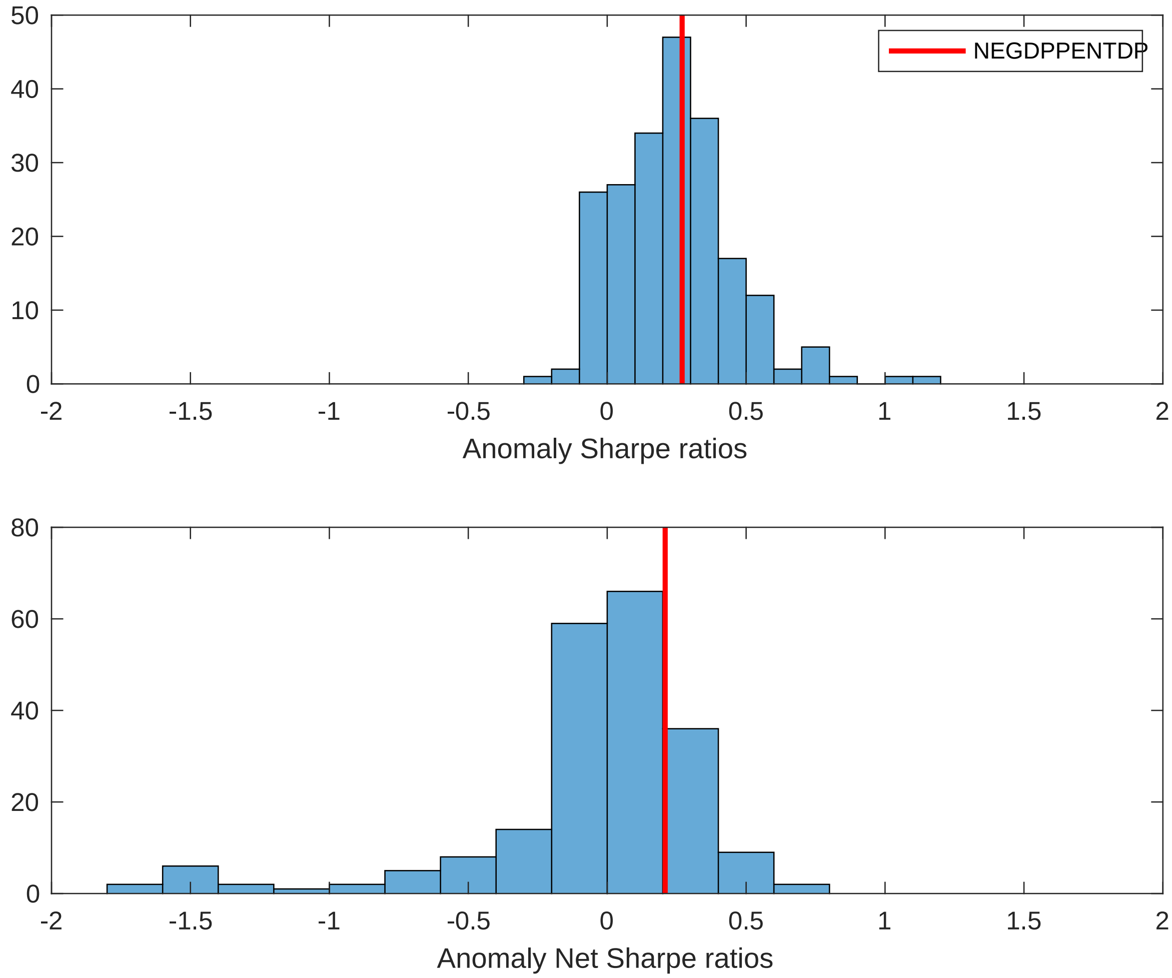


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

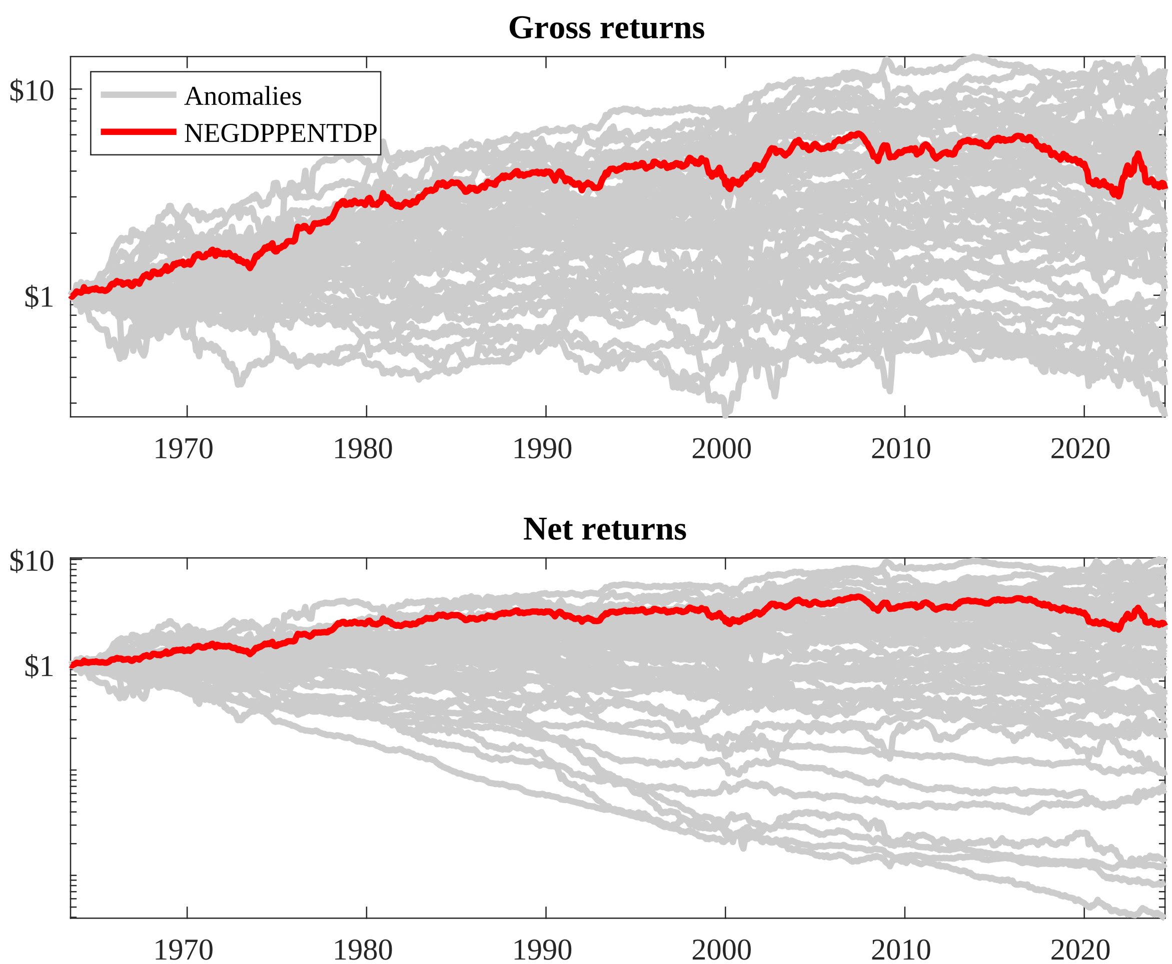
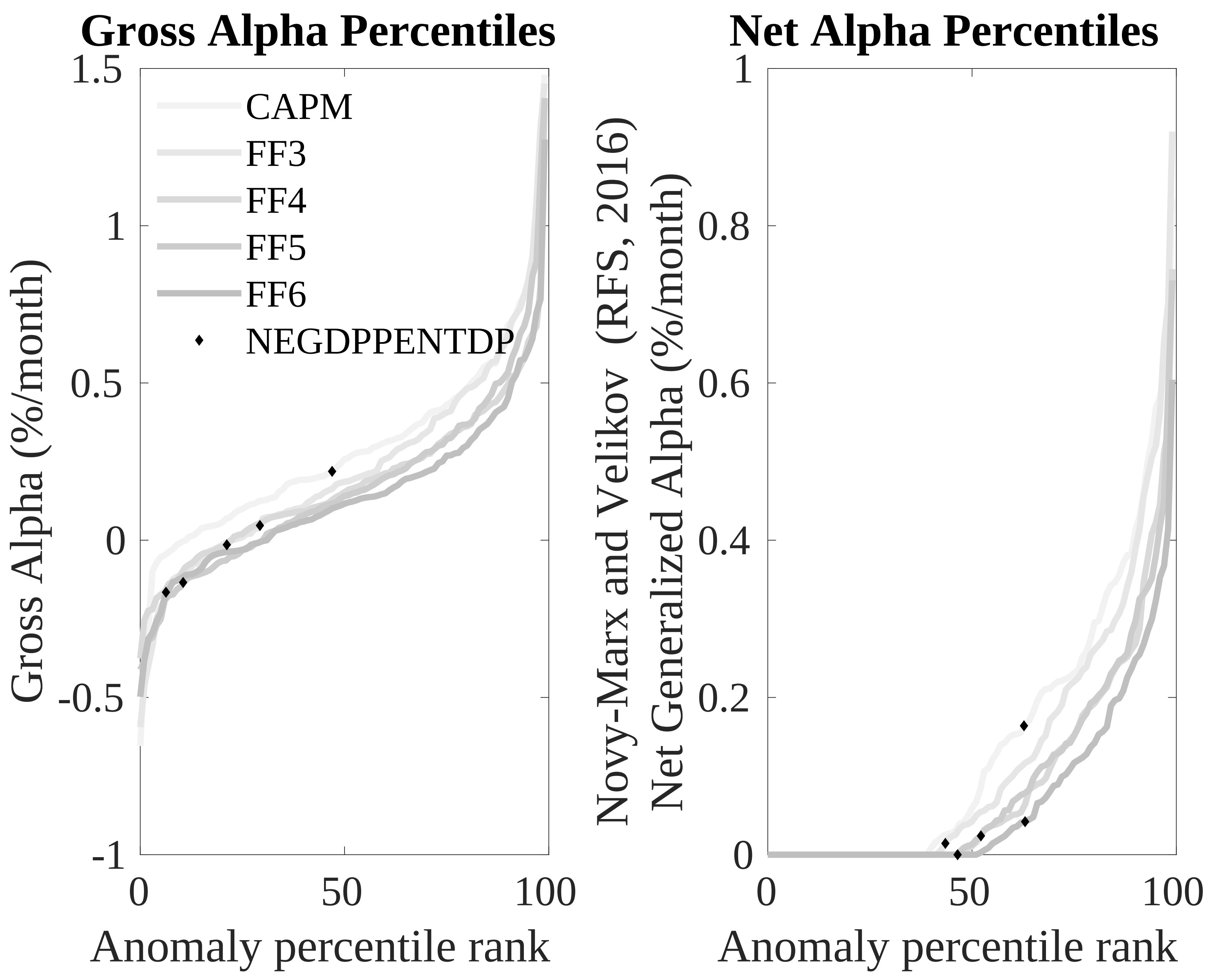
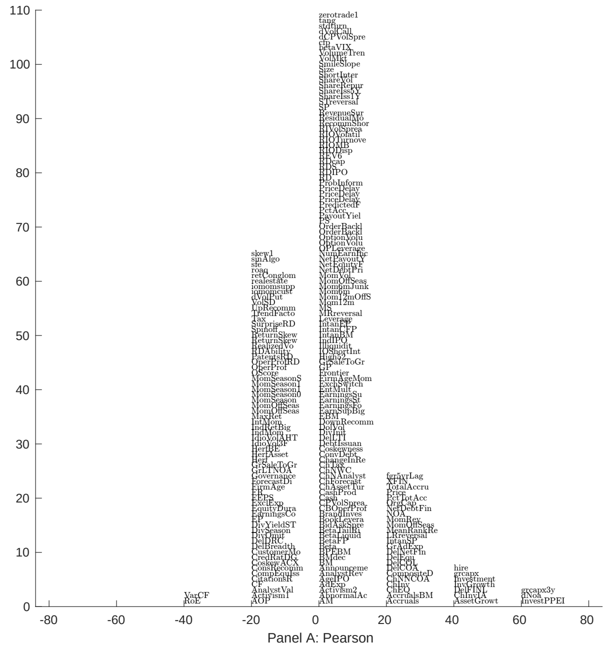


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Gross Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDPPENTDP
Anomaly percentile rank
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank



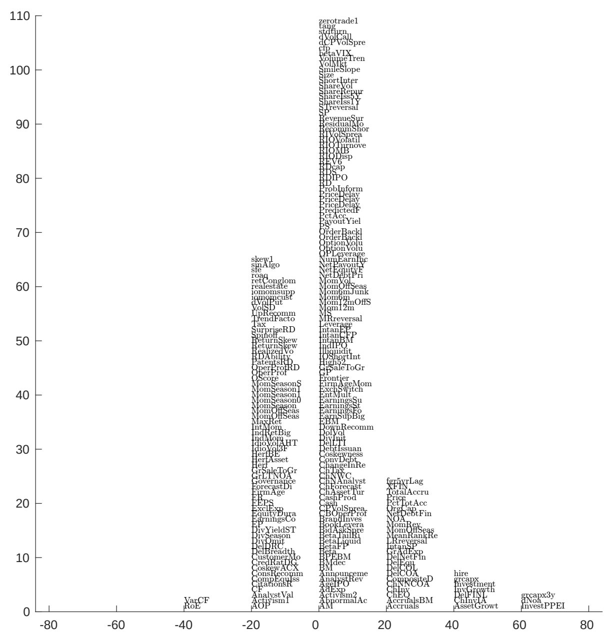


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CRI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

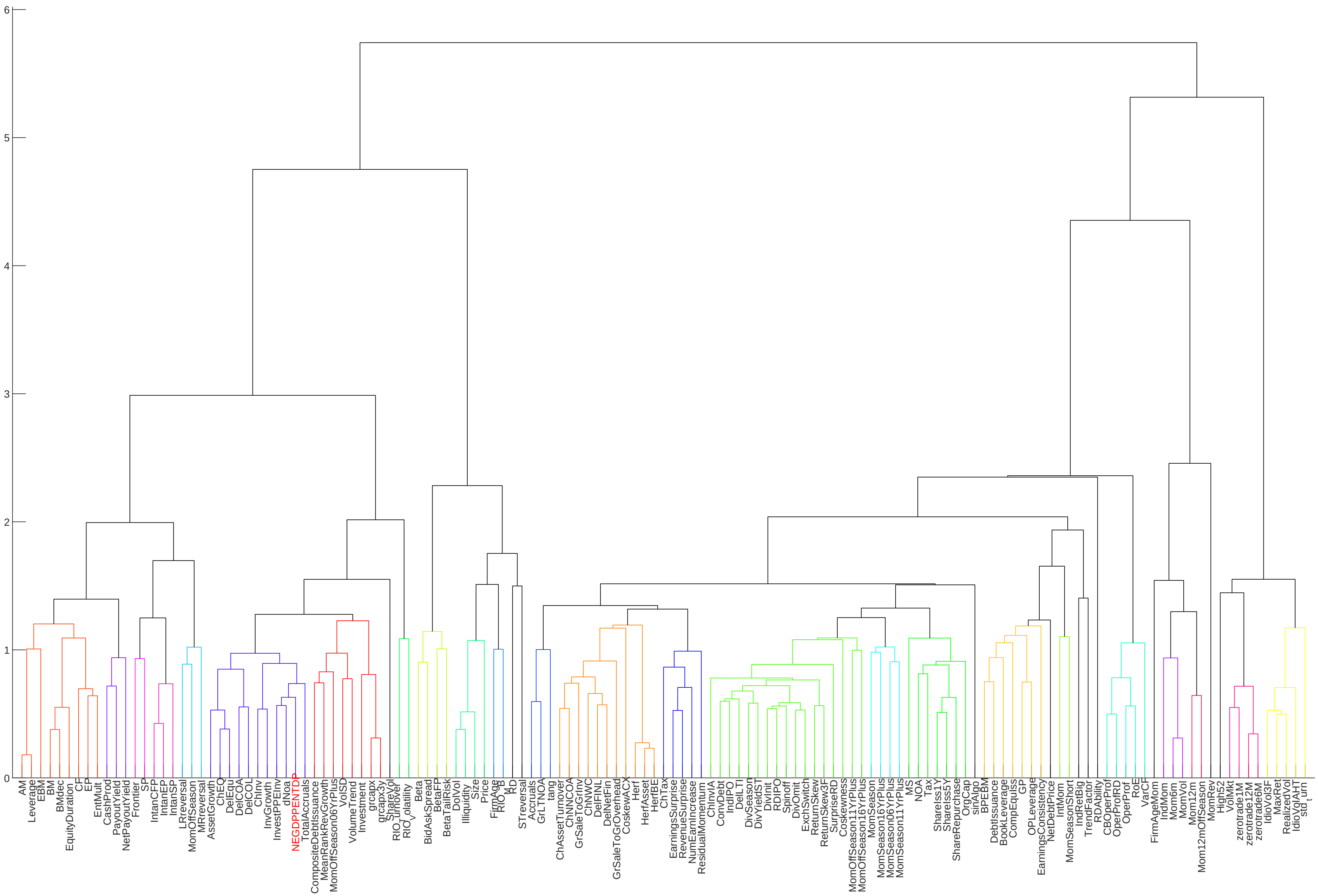


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

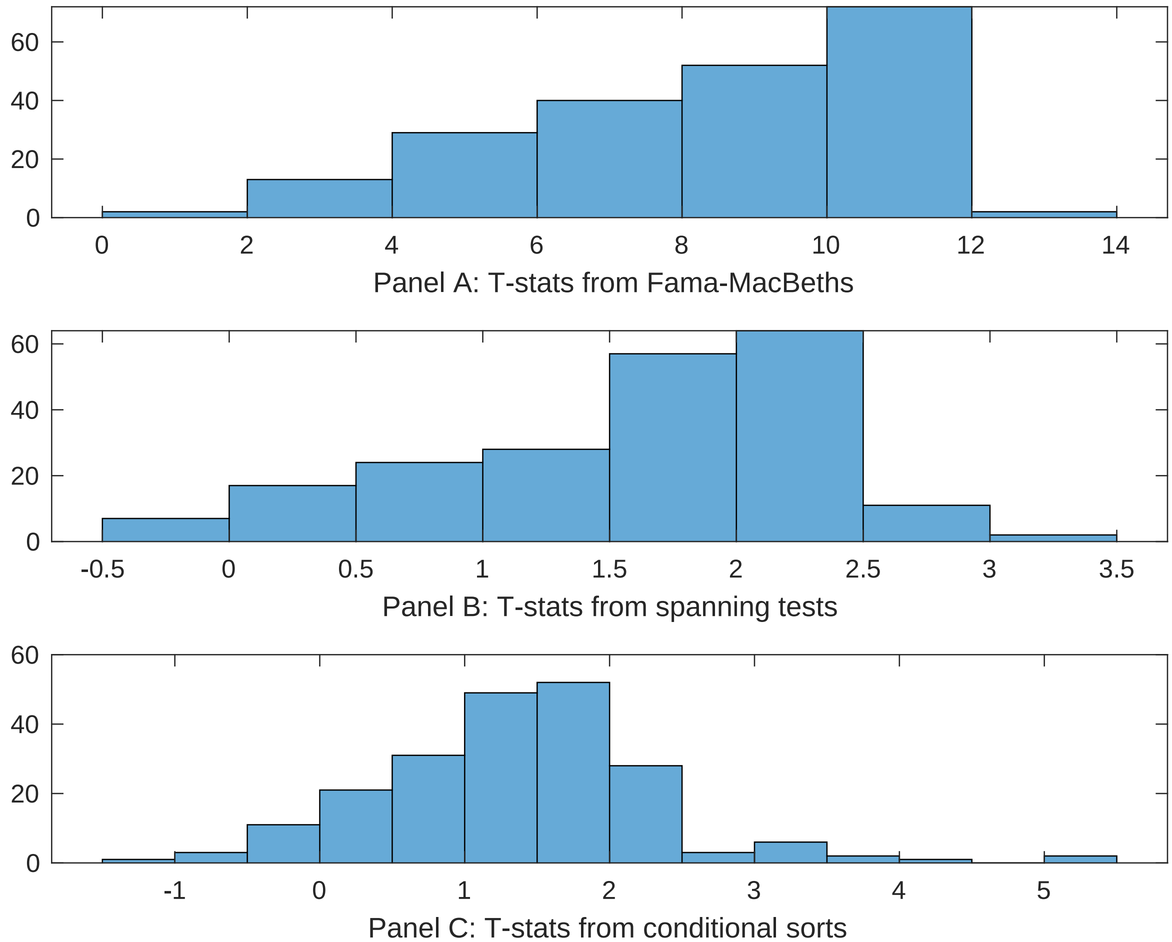


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CRI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRI} CRI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRI. Stocks are finally grouped into five CRI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRI} CRI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14	0.14	0.13	0.13	0.14	0.13	0.14
	[6.16]	[6.58]	[6.12]	[6.14]	[6.39]	[5.84]	[6.12]
CRI	0.40	0.47	0.39	0.51	0.37	0.54	0.23
	[6.23]	[7.50]	[6.59]	[8.49]	[6.37]	[9.78]	[3.19]
Anomaly 1	0.11						-0.10
	[5.48]						[-0.04]
Anomaly 2		0.86					0.25
		[5.70]					[1.11]
Anomaly 3			0.10				0.23
			[6.77]				[1.25]
Anomaly 4				0.35			0.13
				[4.61]			[1.03]
Anomaly 5					0.82		0.44
					[6.70]		[2.54]
Anomaly 6						0.59	0.90
						[4.23]	[0.66]
# months	732	727	721	727	732	713	708
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are change in ppe and inv/assets, Change in capex (three years), change in net operating assets, Change in capex (two years), Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.16	-0.13	-0.22	-0.15	-0.17	-0.08	-0.15
	[-2.22]	[-1.84]	[-3.00]	[-2.06]	[-2.13]	[-1.07]	[-2.16]
Anomaly 1	43.78						26.51
	[13.09]						[7.18]
Anomaly 2		39.95					23.87
		[11.05]					[3.89]
Anomaly 3			40.13				19.73
			[9.52]				[4.36]
Anomaly 4				35.66			1.20
				[9.68]			[0.20]
Anomaly 5					22.83		4.80
					[4.89]		[1.02]
Anomaly 6						28.90	7.18
						[6.79]	[1.74]
mkt	8.23	8.02	8.96	8.32	8.79	9.22	8.11
	[4.82]	[4.56]	[5.00]	[4.65]	[4.71]	[4.97]	[4.97]
smb	19.47	13.29	21.65	16.76	19.39	22.69	15.59
	[7.88]	[5.04]	[8.36]	[6.39]	[7.11]	[8.45]	[6.20]
hml	7.26	3.89	8.88	5.07	11.30	6.58	0.67
	[2.18]	[1.12]	[2.55]	[1.44]	[3.13]	[1.79]	[0.21]
rmw	14.60	11.71	15.18	11.11	14.53	15.40	13.08
	[4.38]	[3.41]	[4.34]	[3.17]	[3.99]	[4.26]	[4.09]
cma	35.81	48.68	39.48	50.51	41.86	44.45	9.18
	[6.52]	[9.12]	[6.60]	[9.28]	[5.41]	[6.87]	[1.26]
umd	4.05	1.76	3.28	1.49	4.84	2.41	1.76
	[2.39]	[1.00]	[1.84]	[0.83]	[2.60]	[1.30]	[1.06]
# months	732	728	732	728	732	713	709
$\bar{R}^2$(%)	52	49	47	47	43	45	57

Panel A: Average rank

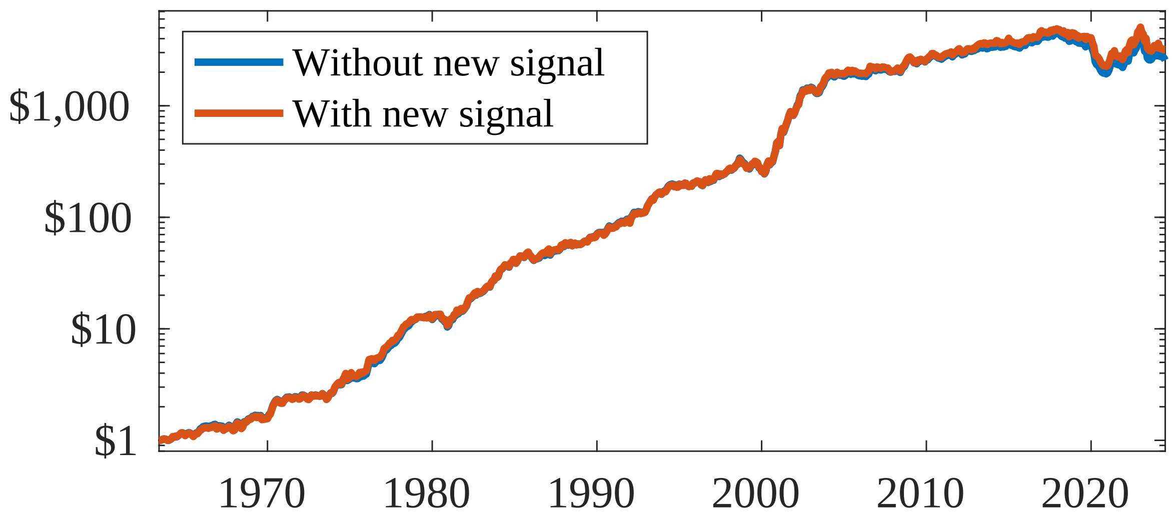


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CRI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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